

Interactions and Correlations in Condensed Matter

Diagrammatic Monte Carlo - Part II

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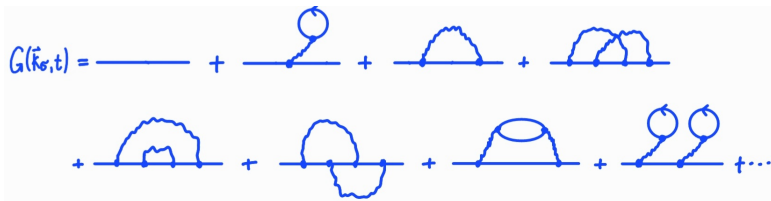
Green's function for the polaron problem

Polaron Hamiltonian

$$\hat{H}_{pol} = \sum_{\mathbf{k}} \varepsilon_{\mathbf{k}} \hat{a}_{\mathbf{k}}^{\dagger} \hat{a}_{\mathbf{k}} + \sum_{\mathbf{q}} \hbar \omega_{\mathbf{q}} \hat{b}_{\mathbf{q}}^{\dagger} \hat{b}_{\mathbf{q}} + \sum_{\mathbf{k}, \mathbf{q}} M_{\mathbf{q}} \hat{a}_{\mathbf{k}+\mathbf{q}}^{\dagger} \hat{a}_{\mathbf{k}} \left(\hat{b}_{\mathbf{q}} + \hat{b}_{-\mathbf{q}}^{\dagger} \right)$$

Feynman diagrams for this system consist of:

- non-interacting electron Green's function $G^{(0)}(\mathbf{k}, t)$ or $G^{(0)}(\mathbf{k}, \omega)$
- non-interacting phonon Green's function $D^{(0)}(\mathbf{q}, t)$ or $D^{(0)}(\mathbf{q}, \omega)$
- electron-phonon matrix element $M_{\mathbf{q}}$
- occupation number $n_F(\varepsilon_{\mathbf{k}})$



Green's function for the polaron problem

In the polaron problem, we are usually only interested in a single electron interacting with a bath of phonons:

$$G(\vec{k}, t) = \text{---} + \cancel{\text{---} \circ \text{---}} + \text{---} \text{---} + \text{---} \text{---} \text{---} + \text{---} \text{---} \text{---} \text{---} + \text{---} \text{---} \text{---} \text{---} + \cancel{\text{---} \text{---} \text{---}} + \cancel{\text{---} \text{---} \text{---} \text{---}} + \dots$$

This gives the final diagrammatic expansion for the electron Green's function:

$$G(\vec{k}, t) = \text{---} + \text{---} \text{---} + \text{---} \text{---} \text{---} + \text{---} \text{---} \text{---} \text{---} + \text{---} \text{---} \text{---} \text{---} \text{---} + \dots$$

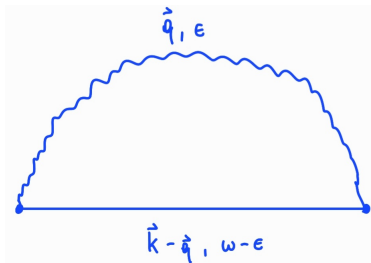
Polaron energy from Green's functions

The polaron energy (quasiparticle energy) can be obtained in two equivalent ways:

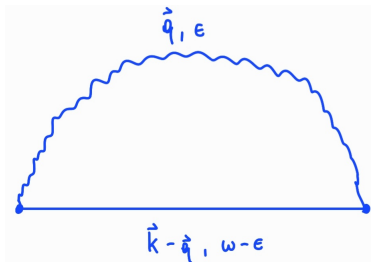
- poles of the Green's function $G(\mathbf{k}, \omega)$
- real part of the self-energy $E_{\mathbf{k}} = \varepsilon_{\mathbf{k}} + \text{Re} [\Sigma(\mathbf{k}, E_{\mathbf{k}}/\hbar)]$

Analytically it is favourable to calculate the self-energy $\implies$ less diagrams to evaluate!

Goal: Approximate self-energy to lowest order and calculate the polaron energy. Compare with ordinary perturbation theory.



Polaron energy from Green's functions



Let's use our Feynman rules to translate this diagram into an algebraic equation:

- one electron GF: $G^{(0)}(\mathbf{k} - \mathbf{q}, \omega - \epsilon)$
- one phonon GF: $D^{(0)}(\mathbf{q}, \epsilon)$
- two vertex factors: $M_{-\mathbf{q}}M_{\mathbf{q}} = |M_{\mathbf{q}}|^2$
- a factor of $\frac{i}{\hbar^2}$
- integrate/sum over internal degrees of freedom $\mathbf{q}$ and ϵ

$$\Sigma(\mathbf{k}, \omega) = \frac{i}{\hbar^2} \sum_{\mathbf{q}} \int_{-\infty}^{\infty} \frac{d\epsilon}{2\pi} |M_{\mathbf{q}}|^2 G^{(0)}(\mathbf{k} - \mathbf{q}, \omega - \epsilon) D^{(0)}(\mathbf{q}, \epsilon)$$

Polaron energy from Green's functions

$$\begin{aligned}\Sigma(\mathbf{k}, \omega) &= \frac{i}{\hbar^2} \sum_{\mathbf{q}} \int_{-\infty}^{\infty} \frac{d\epsilon}{2\pi} |M_{\mathbf{q}}|^2 G^{(0)}(\mathbf{k} - \mathbf{q}, \omega - \epsilon) D^{(0)}(\mathbf{q}, \epsilon) \\ &= \frac{i}{\hbar^2} \sum_{\mathbf{q}} |M_{\mathbf{q}}|^2 \int_{-\infty}^{\infty} \frac{d\epsilon}{2\pi} \frac{1}{\omega - \epsilon - \frac{\epsilon_{\mathbf{k}-\mathbf{q}}}{\hbar} + i\delta} \left\{ \frac{1}{\epsilon - \omega_{\mathbf{q}} + i\delta} - \frac{1}{\epsilon + \omega_{\mathbf{q}} - i\delta} \right\} \\ &= \frac{i}{\hbar^2} \sum_{\mathbf{q}} |M_{\mathbf{q}}|^2 \frac{-i}{\omega - \omega_{\mathbf{q}} - \frac{\epsilon_{\mathbf{k}-\mathbf{q}}}{\hbar} + i\delta}\end{aligned}$$

Taking only the real part of this expression and putting in the corresponding quantities for the Fröhlich Hamiltonian as well as using the polaronic units

$$\hbar = \omega_{LO} = 2m = 1:$$

$$\text{Re} [\Sigma(\mathbf{k}, \omega)] = \sum_{\mathbf{q}} \frac{4\pi\alpha}{V} \frac{1}{q^2} \frac{1}{\omega - 1 - (\mathbf{k} - \mathbf{q})^2}$$

Polaron energy from Green's functions

The quasiparticle energy can now be calculated by the following equation

$$E_{\mathbf{k}} = \varepsilon_{\mathbf{k}} + \text{Re} [\Sigma(\mathbf{k}, E_{\mathbf{k}})] = k^2 + \sum_{\mathbf{q}} \frac{4\pi\alpha}{V} \frac{1}{q^2} \frac{1}{E_{\mathbf{k}} - 1 - (\mathbf{k} - \mathbf{q})^2}$$

To approximate a solution, let's use the unperturbed energy $\varepsilon_{\mathbf{k}} = k^2$ in the self-energy instead of $E_{\mathbf{k}}$:

$$\begin{aligned} E_{\mathbf{k}} = \varepsilon_{\mathbf{k}} + \text{Re} [\Sigma(\mathbf{k}, k^2)] &= k^2 + \sum_{\mathbf{q}} \frac{4\pi\alpha}{V} \frac{1}{q^2} \frac{1}{k^2 - 1 - (\mathbf{k} - \mathbf{q})^2} \\ &= k^2 - \alpha - k^2 \frac{\alpha}{6} \\ &= -\alpha + k^2 \left(1 - \frac{\alpha}{6}\right) \end{aligned}$$

This is nothing but ordinary 2nd-order Rayleigh-Schrödinger perturbation theory!

How can we improve this result and go beyond ordinary perturbation theory? $\implies$ **Diagrammatic Monte Carlo**

DiagMC: Summary (recap)

DiagMC allows us to calculate quantities which can be written in a diagrammatic series of the form

$$Q(\{y\}) = \sum_{n=0}^{\infty} \sum_{\xi_n} \int \cdots \int \mathcal{D}_n^{\xi_n}(\{y\}; x_1, \dots, x_n) dx_1 \dots dx_n$$

The basic ingredients of any DiagMC code are therefore:

- **diagrammatic expansion** of $Q(\{y\})$ and rules to translate between the numeric and diagrammatic expressions of $\mathcal{D}_n^{\xi_n}(\{y\}; x_1, \dots, x_n)$
- **updates** to propose a new diagram given a current diagram
 - ▶ theoretically we have to be able to generate and reach any diagram in the expansion $\implies$ ergodicity!
 - ▶ Metropolis-Hasting algorithm requires detailed balance $\implies$ each update needs an inverse!
- **measurements** to be performed during the simulation
 - ▶ histogram for $Q(\{y\})$
 - ▶ diagram statistics
 - ▶ ...

DiagMC: General strategy (recap)

The general strategy (MC loop) of our DiagMC examples will all follow the same basic scheme:

1. start from some initial diagram (usually some 0th-order diagram)
2. warm up the underlying Markov chain (until stationary distribution is reached)
 - 2.1. choose an update U_i from the set of updates $\mathcal{U}$
 - 2.2. propose a new diagram using U_i
 - 2.3. accept or reject the new diagram with probability P_{acc} given by the Metropolis-Hasting algorithm
3. run the main simulation
 - 3.1. choose an update U_i from the set of updates $\mathcal{U}$
 - 3.2. propose a new diagram using U_i
 - 3.3. accept or reject the new diagram with probability P_{acc} given by the Metropolis-Hasting algorithm
 - 3.4. perform measurements
4. output results/post processing

Imaginary time vs. real time

DiagMC uses the value of the diagrams as statistical weights. This means that the numerical values of the diagrams $\mathcal{D}_n^{\xi_n}(\{y\}; x_1, \dots, x_n)$ appear in the acceptance probability of the Metropolis-Hastings algorithm:

- **Class I update:**

$$P_{acc} = \min \left\{ 1, \frac{\mathcal{D}_n^{\xi_n}(\{y\}; x_1, \dots, x'_i, \dots, x_n) p(x_i | x'_i)}{\mathcal{D}_n^{\xi_n}(\{y\}; x_1, \dots, x_i, \dots, x_n) p(x'_i | x_i)} \right\}$$

- **Class II update:**

$$P_{acc} = \min \left\{ 1, \frac{p_B}{p_A} \frac{\mathcal{D}_{n+m}^{\xi_{n+m}}(\{y\}; x_1, \dots, x_n, \dots, x_{n+m})}{\mathcal{D}_n^{\xi_n}(\{y\}; x_1, \dots, x_n) p(x_{n+1}, \dots, x_{n+m})} \right\}$$

If we work in real time (or real frequencies), $\mathcal{D}_n^{\xi_n}(\{y\}; x_1, \dots, x_i, \dots, x_n)$ are in general complex quantities.

How to deal with this?

Imaginary time vs. real time

$\Rightarrow$ use $|\mathcal{D}_n^{\xi}(\{y\}; x_1, \dots, x_i, \dots, x_n)|$ (absolute value) as the statistical weight

To simplify the following analysis, we can consider a simple integral with a non-sign definite function $w(x)$ (which we would like to use as a distribution):

$$\begin{aligned}\frac{\int dx w(x) f(x)}{\int dx w(x)} &= \frac{\int dx |w(x)| \operatorname{sgn}[w(x)] f(x)}{\int dx |w(x)| \operatorname{sgn}[w(x)]} \\&= \frac{\int dx |w(x)| \operatorname{sgn}[w(x)] f(x)}{\int dx |w(x)|} \frac{\int dx |w(x)|}{\int dx |w(x)| \operatorname{sgn}[w(x)]} \\&= \frac{\langle \operatorname{sgn}[w(x)] f(x) \rangle_{|w(x)|}}{\langle \operatorname{sgn}[w(x)] \rangle_{|w(x)|}}\end{aligned}$$

$\int dx |w(x)| / \int dx |w(x)|$: needed for normalization

If the average sign $\langle \operatorname{sgn}[w(x)] \rangle_{|w(x)|}$ is close to zero $\Rightarrow$ sign problem!

Using the **imaginary time (Matsubara)** representation, we can avoid the sign problem for some of our examples (not in general!), e.g.: Green's functions in imaginary time become real, not complex anymore.

Example 1: Free propagator

We start with the simplest example, calculating a Matsubara function of a non-interacting particle:

$$\hat{H} = \varepsilon \hat{c}^\dagger \hat{c}$$

$$Q(\{\tau\}) = e^{\mu\tau} G^{(0)}(\tau) = e^{\mu\tau} \langle \emptyset | \hat{c}(\tau) \hat{c}^\dagger(0) | \emptyset \rangle = e^{-(\varepsilon-\mu)\tau} = \mathcal{D}_0(\tau)$$

- $\tau \geq 0$
- ε is the energy of the particle
- μ is an artificial chemical potential (simulation parameter)
- τ_{max} is the max. allowed τ value (simulation parameter)

Diagrammatic series:

- only 0th-order diagram

$$G^{(0)}(\tau) = \frac{\quad}{0 \quad \tau} = e^{-(\varepsilon-\mu)\tau}$$

Example 1: Free propagator

$$Q(\{\tau\}) = e^{\mu\tau} G^{(0)}(\tau) = e^{\mu\tau} \langle \emptyset | \hat{c}(\tau) \hat{c}^\dagger(0) | \emptyset \rangle = e^{-(\varepsilon - \mu)\tau} = \mathcal{D}_0(\tau)$$

Updates:

- change- τ update:

$$\mathcal{D}_0(\tau) = \text{---} \underset{0}{\text{---}} \underset{\tau}{\text{---}} \longleftrightarrow \text{---} \underset{0}{\text{---}} \underset{\tau'}{\text{---}} = \mathcal{D}_0(\tau')$$

- ▶ draw new length τ' uniformly from $\mathcal{U}(0, \tau_{\max})$
- ▶ acceptance probability:

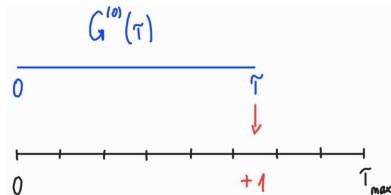
$$P_{\text{acc}} = \min \left\{ 1, \frac{\mathcal{D}_0(\tau')}{\mathcal{D}_0(\tau)} \right\} = \min \left\{ 1, e^{-(\varepsilon - \mu)(\tau' - \tau)} \right\}$$

Example 1: Free propagator

$$Q(\{\tau\}) = e^{\mu\tau} G^{(0)}(\tau) = e^{\mu\tau} \langle \emptyset | \hat{c}(\tau) \hat{c}^\dagger(0) | \emptyset \rangle = e^{-(\varepsilon - \mu)\tau} = \mathcal{D}_0(\tau)$$

Measurements:

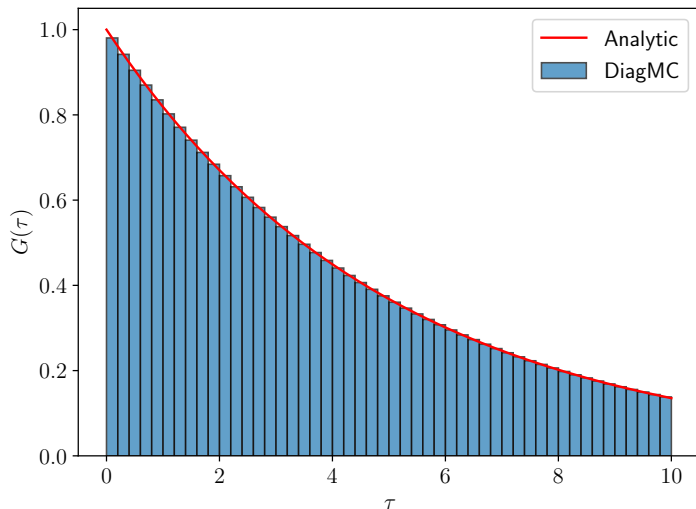
- histogram $H(\tau_i)$ for $G^{(0)}(\tau)$
 - ▶ M bins β_i of size $\Delta\tau_i$ from 0 to τ_{max}



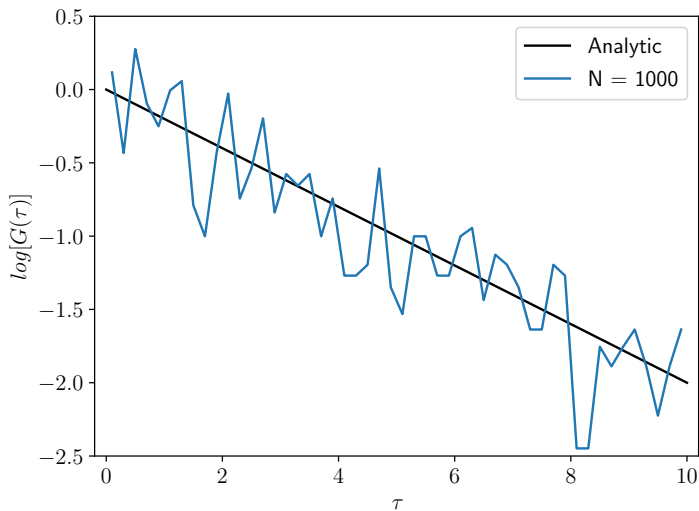
- ▶ normalization:

$$I = \int_0^{\tau_{max}} d\tau Q(\{\tau\}), \quad S = \sum_{i=1}^M \tilde{H}(\tau_i) \Delta\tau_i \quad \Rightarrow \quad H(\tau_i) = \tilde{H}(\tau_i) \frac{I}{S}$$

Example 1: Free propagator

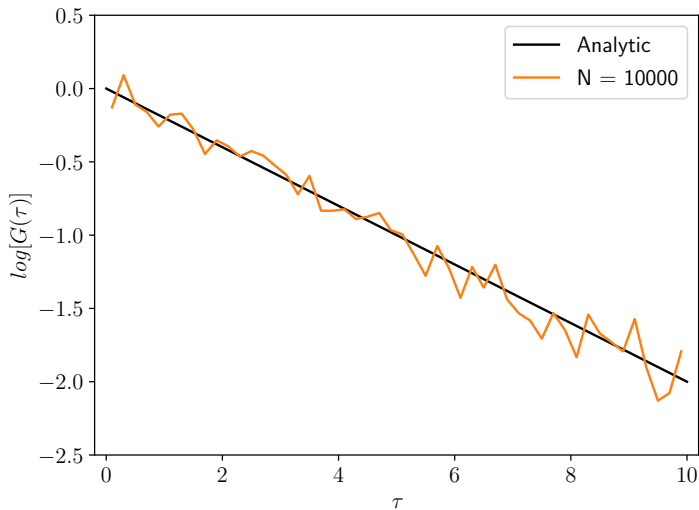


Example 1: Free propagator



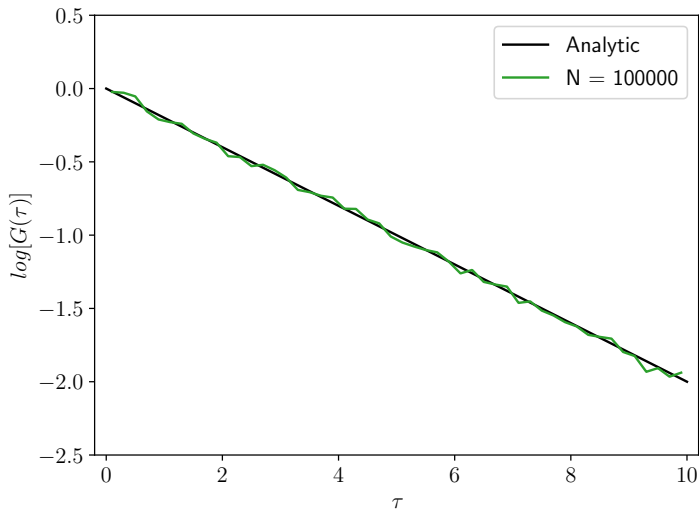
N = number of DiagMC steps

Example 1: Free propagator



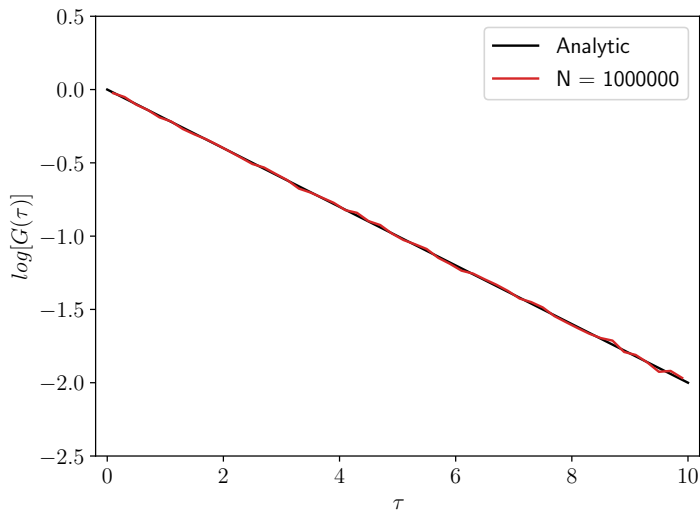
N = number of DiagMC steps

Example 1: Free propagator



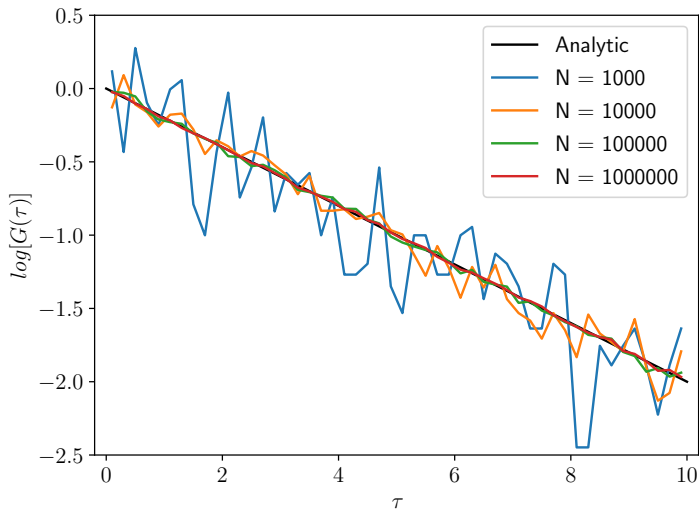
N = number of DiagMC steps

Example 1: Free propagator



N = number of DiagMC steps

Example 1: Free propagator



N = number of DiagMC steps

Example 2: (Attractive) potential

Next we add a constant potential to our non-interacting particle. Since this results in a constant energy shift, it is still a trivial problem:

$$\hat{H} = \varepsilon \hat{c}^\dagger \hat{c} + \Delta \hat{c}^\dagger \hat{c} = (\varepsilon + \Delta) \hat{c}^\dagger \hat{c}$$

$$Q(\{\tau\}) = e^{\mu\tau} G^{(0)}(\tau) = e^{\mu\tau} \langle \emptyset | \hat{c}(\tau) \hat{c}^\dagger(0) | \emptyset \rangle = e^{-(\varepsilon - \mu + \Delta)\tau} = \mathcal{D}_0(\tau)$$

- $\tau \geq 0$
- ε is the energy of the particle
- Δ is the strength of the potential (we assume it to be attractive for the time being, i.e. $\Delta \leq 0$)
- μ is an artificial chemical potential (simulation parameter)
- τ_{max} is the max. allowed τ value (simulation parameter)

$\Rightarrow$ we can approach this problem with 3 different DiagMC strategies:

Example 2: (Attractive) potential

$$\hat{H} = (\varepsilon - \mu)\hat{c}^\dagger\hat{c} + \Delta\hat{c}^\dagger\hat{c} = (\varepsilon - \mu + \Delta)\hat{c}^\dagger\hat{c}$$

$$Q(\{\tau\}) = e^{\mu\tau} G^{(0)}(\tau) = e^{\mu\tau} \langle \emptyset | \hat{c}(\tau) \hat{c}^\dagger(0) | \emptyset \rangle = e^{-(\varepsilon - \mu + \Delta)\tau} = \mathcal{D}_0(\tau)$$

1. only 0th-order diagram:

$$G^{(0)}(\tau) = \frac{\text{---}}{0 \quad \tau} = e^{-(\varepsilon - \mu + \Delta)\tau}$$

2. full diagrammatic expansion with potential as interaction term:

$$G(\tau) = \text{---} + \text{---} \times \text{---} + \text{---} \times \times \text{---} + \dots$$

$$+ \text{---} \times \times \times \text{---} + \text{---} \times \times \times \times \text{---} + \text{---} \times \times \times \times \times \text{---} + \dots$$

$$e^{\mu\tau} G(\tau) = \sum_{n=0}^{\infty} (-1)^n \int_0^{\tau} d\tau_n \dots \int_0^{\tau_{i+1}} d\tau_i \dots \int_0^{\tau_2} d\tau_1 e^{-(\varepsilon - \mu)\tau_1} \Delta \dots \Delta e^{-(\varepsilon - \mu)(\tau - \tau_n)}$$

Sign problem only if $\Delta > 0$. For $\Delta < 0$, $(-1)^n \Delta^n$ always +

Example 2: (Attractive) potential

3. full diagrammatic expansion without integrations:

$$e^{\mu\tau} G(\tau) = \sum_{n=0}^{\infty} (-1)^n \int_0^{\tau} d\tau_n \cdots \int_0^{\tau_{i+1}} d\tau_i \cdots \int_0^{\tau_2} d\tau_1 e^{-(\varepsilon-\mu)\tau_1} \Delta \dots \Delta e^{-(\varepsilon-\mu)(\tau-\tau_n)}$$

Exponential functions cancel out, only the last remains

$$\begin{aligned} &= e^{-(\varepsilon-\mu)\tau} \sum_{n=0}^{\infty} (-1)^n \Delta^n \int_0^{\tau} d\tau_n \cdots \int_0^{\tau_{i+1}} d\tau_i \cdots \int_0^{\tau_2} d\tau_1 \\ &= e^{-(\varepsilon-\mu)\tau} \sum_{n=0}^{\infty} (-1)^n \Delta^n \frac{\tau^n}{n!} \quad (\text{solvable with DiagMC}) \\ &= e^{-(\varepsilon-\mu)\tau} e^{-\Delta\tau} = e^{-(\varepsilon-\mu+\Delta)\tau} \end{aligned}$$

In the following, we focus on the 2. approach, since it has a similar structure to more complicated examples treated later.

Example 2: (Attractive) potential

Diagrammatic series: $\mathcal{D}_n(\tau) = \Delta^n e^{-(\varepsilon - \mu)\tau}$

$$G(\tau) = \text{---} + \text{---} \times \text{---} + \text{---} \times \times \text{---} + \dots$$

$$+ \text{---} \times \times \times \text{---} + \text{---} \times \times \times \times \text{---} + \text{---} \times \times \times \times \times \text{---} + \dots$$

Updates:

- change- τ update:

$$\mathcal{D}_h(\tau) = \text{---} \times \times \text{---} \xleftrightarrow{\quad} \text{---} \times \times \text{---} = \mathcal{D}_h(\tau')$$

0 τ 0 τ'

- ▶ draw new length τ' uniformly from $\mathcal{U}(\tau_L, \tau_{max})$ where τ_L is the time point of the last (rightmost) vertex in the diagram
- ▶ acceptance probability:

$$P_{acc} = \min \left\{ 1, \frac{\mathcal{D}_n(\tau')}{\mathcal{D}_n(\tau)} \right\} = \min \left\{ 1, e^{-(\varepsilon - \mu)(\tau' - \tau)} \right\}$$

Example 2: (Attractive) potential

- add-vertex update:

$$\mathcal{D}_n(\tau) = \begin{array}{c} \times \quad \times \\ \hline 0 \quad \tau_L \quad \tau_R \quad \tau \end{array} \longleftrightarrow \begin{array}{c} \times \quad \times \quad \times \\ \hline 0 \quad \tau_L \quad \tau_{n+1} \quad \tau_R \quad \tau \end{array} = \mathcal{D}_{n+1}(\tau)$$

- ▶ choose a random particle propagator of length $\tau_R - \tau_L$
- ▶ draw a time point τ_{n+1} uniformly from $\mathcal{U}(\tau_L, \tau_R)$
- ▶ acceptance probability:

$$\begin{aligned} P_{acc} &= \min \left\{ 1, \frac{p_{rem}}{p_{add}} \frac{\mathcal{D}_{n+1}(\tau)}{\mathcal{D}_n(\tau) \mathcal{U}(\tau_L, \tau_R)} \right\} = \min \left\{ 1, \frac{p_{rem}}{p_{add}} \Delta(\tau_R - \tau_L) \right\} \\ &= \min \left\{ 1, \frac{p_{rem}}{p_{add}} \frac{n+1}{n+1} \Delta(\tau_R - \tau_L) \right\} \end{aligned}$$

$\mathcal{U}(\tau_L, \tau_R)$: $p(x_{n+1}, \dots, x_{n+m})$ in the general formula

$$\mathcal{D}_{n+1}(\tau)/\mathcal{D}_n(\tau) = \Delta \quad \mathcal{U}(\tau_L, \tau_R) = 1/(\tau_R - \tau_L)$$

p split into a multiplicative factor $(n+1)$ and probabilities P_{add} and P_{rem} with which the add and remove update are chosen. Needed for detailed balance.

Example 2: (Attractive) potential

- remove-vertex update:

$$\mathcal{D}_n(\tau) = \overbrace{0 \quad \tau_L \quad \tau_R \quad \tau}^{\text{---}} \longleftrightarrow \overbrace{0 \quad \tau_L \quad \tau_{n+1} \quad \tau_R \quad \tau}^{\text{---}} = \mathcal{D}_{n+1}(\tau)$$

- is the inverse update to add-vertex
- choose a random vertex uniformly
- acceptance probability:

$$\begin{aligned} P_{acc} &= \min \left\{ 1, \frac{p_{add}}{p_{rem}} \frac{\mathcal{D}_{n-1}(\tau) \mathcal{U}(\tau_L, \tau_R)}{\mathcal{D}_n(\tau)} \right\} \\ &= \min \left\{ 1, \frac{p_{add}}{p_{rem}} \frac{n}{n} \frac{1}{\Delta(\tau_R - \tau_L)} \right\} \end{aligned}$$

where τ_L and τ_R are the time points of the vertices to the left and to the right of the vertex to be removed.

Example 2: (Attractive) potential

Measurements:

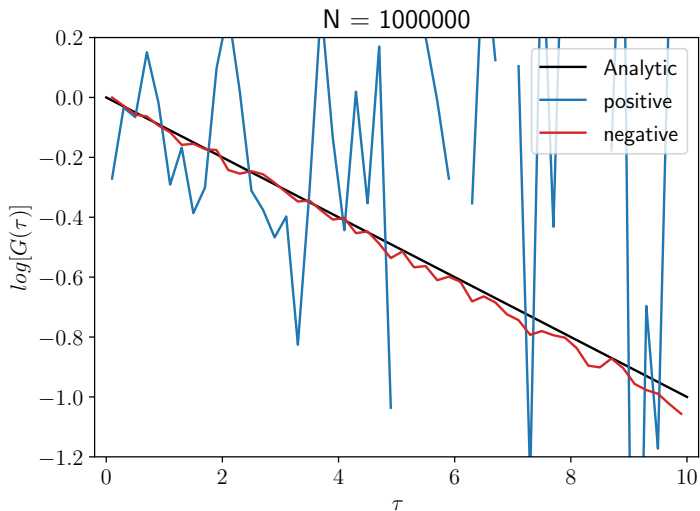
- histogram $H(\tau_i)$ for $G^{(0)}(\tau)$
 - ▶ procedure is exactly the same as before
 - ▶ normalization: usually it is not possible to compute the normalization factor analytically; trick: use 0th-order diagram (easy to compute analytically). Then, add N_0 counting how often 0th-order diagrams are generated.

$$I_0 = \int_0^{\tau_{\max}} d\tau \mathcal{D}_0(\tau) = \int_0^{\tau_{\max}} d\tau e^{-(\varepsilon - \mu)\tau} = \frac{1 - e^{-(\varepsilon - \mu)\tau_{\max}}}{\varepsilon - \mu}$$

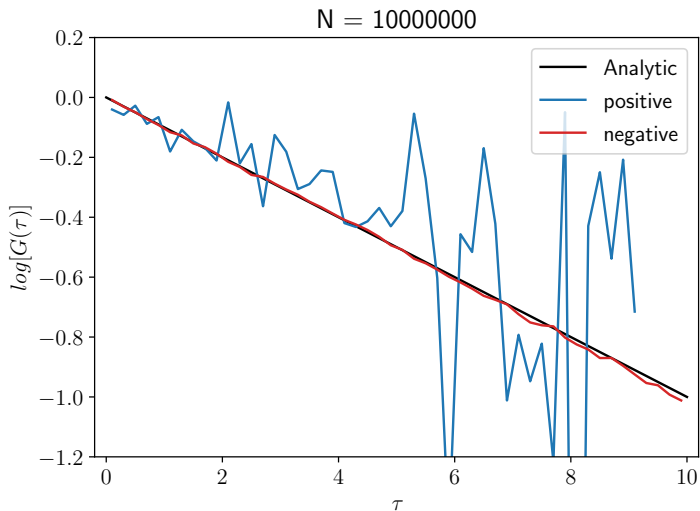
$$\implies H(\tau_i) = \frac{\tilde{H}(\tau_i)}{\Delta\tau_i} \frac{I_0}{N_0}$$

- update statistics
 - ▶ acceptance-rejection ratio
 - ▶ number of times an update has been used
- diagram statistics
 - ▶ average diagram order
 - ▶ highest order reached
 - ▶ average sign of diagram weights

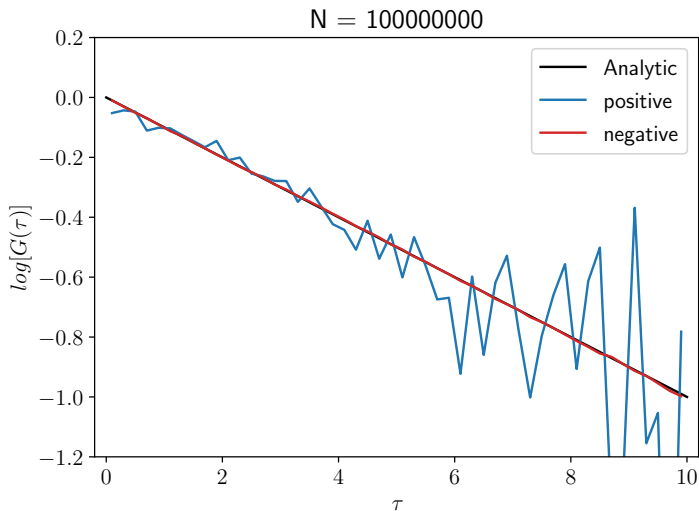
Example 2: (Attractive) potential



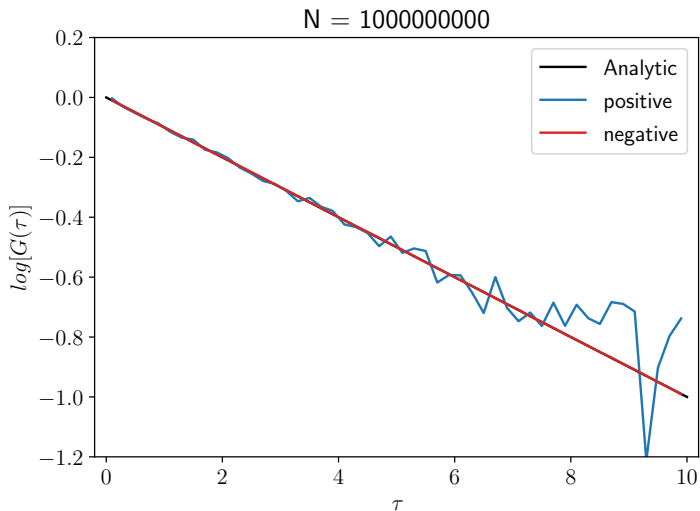
Example 2: (Attractive) potential



Example 2: (Attractive) potential



Example 2: (Attractive) potential



Example 3: Two-Level-System

The next step is to consider a system with two levels and interaction between them:

$$\hat{H} = \frac{\epsilon}{2} \left\{ \hat{c}_1^\dagger \hat{c}_1 - \hat{c}_2^\dagger \hat{c}_2 \right\} + \Delta \left\{ \hat{c}_1^\dagger \hat{c}_2 + \hat{c}_2^\dagger \hat{c}_1 \right\}$$

$$Q(\{i,j,\tau\}) = e^{\mu\tau} G_{ij}(\tau) = e^{\mu\tau} \left\langle \emptyset \left| \hat{a}_i(\tau) \hat{a}_j^\dagger(0) \right| \emptyset \right\rangle$$

- $\tau \geq 0$
- ϵ is the energy gap between the two levels, i.e. $\epsilon_1 = \frac{\epsilon}{2}$ and $\epsilon_2 = -\frac{\epsilon}{2}$
- Δ is the interaction strength (sign problem for $\Delta < 0$)
- μ is an artificial chemical potential (simulation parameter)
- τ_{max} is the max. allowed τ value (simulation parameter)

This problem can still be solved analytically by simply diagonalizing the Hamiltonian with the unitary transformation U :

$$\hat{H} = \begin{pmatrix} \hat{c}_1^\dagger & \hat{c}_2^\dagger \end{pmatrix} \begin{pmatrix} \frac{\epsilon}{2} & \Delta \\ \Delta & -\frac{\epsilon}{2} \end{pmatrix} \begin{pmatrix} \hat{c}_1 \\ \hat{c}_2 \end{pmatrix} = \begin{pmatrix} \hat{c}_1^\dagger & \hat{c}_2^\dagger \end{pmatrix} U \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} U^\dagger \begin{pmatrix} \hat{c}_1 \\ \hat{c}_2 \end{pmatrix}$$

Example 3: Two-Level-System (optional)

Diagrammatic series:

$$G_{ii}(\tau) = \text{---}_0^\tau + \text{---}_0^x \text{---}^x \tau + \text{---}_0^x \text{---}^x \text{---}^x \tau + \dots$$

$$G_{ij}(\tau) = \text{---}_0^x \text{---} \tau + \text{---}_0^x \text{---}^x \text{---} \tau + \text{---}_0^x \text{---}^x \text{---}^x \text{---} \tau + \dots$$

$$\begin{aligned} e^{\mu\tau} G_{ij}(\tau) &= \sum_{n=0}^{\infty} (-1)^n \int_0^\tau d\tau_n \cdots \int_0^{\tau_{i+1}} d\tau_i \cdots \int_0^{\tau_2} d\tau_1 e^{-(\varepsilon_i - \mu)\tau_1} \Delta \dots \Delta e^{-(\varepsilon_j - \mu)(\tau - \tau_n)} \\ &= \sum_{n=0}^{\infty} \int_0^\tau d\tau_n \cdots \int_0^{\tau_{i+1}} d\tau_i \cdots \int_0^{\tau_2} d\tau_1 \mathcal{D}_n(i, j, \tau; \tau_1, \dots, \tau_n) \end{aligned}$$

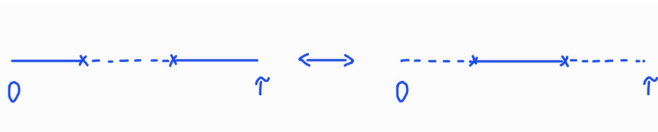
Example 3: Two-Level-System

Updates:

- change- τ update:
 - ▶ depends on the energy of ε_L of the last propagator in the diagram
- add/remove-vertex updates:



- ▶ switches between diagonal G_{ii} and non-diagonal G_{ij} Green's function diagrams
 - ▶ changes all propagators to the right of the added/removed vertex
- flip-states update:
 - ▶ changes the state of all propagators, i.e. $1 \rightarrow 2$ and $2 \rightarrow 1$
 - ▶ switches between different Green's functions, i.e. $G_{11} \leftrightarrow G_{22}$ or $G_{12} \leftrightarrow G_{21}$



Example 3: Two-Level-System

Measurements:

- histogram $H_{ij}(\tau_i)$ for $G_{ij}(\tau)$
 - ▶ procedure is exactly the same as before
 - ▶ normalization: use 0th-order diagram
- update statistics
- diagram statistics

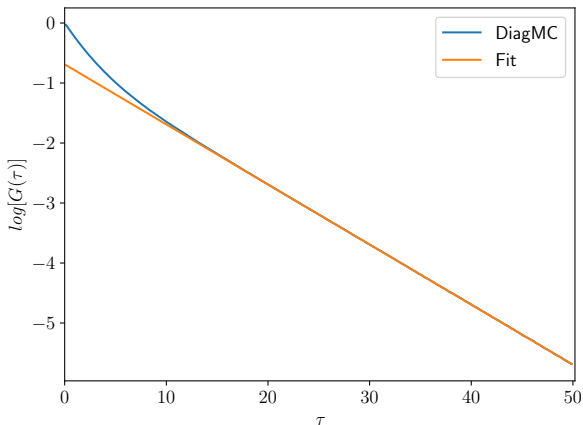
Post processing:

- extract ground state energy E_{GS} and quasiparticle weight $Z_{GS}(i)$ from Green's function:

$$\begin{aligned} e^{\mu\tau} G_{ii}(\tau) &= e^{\mu\tau} \langle \emptyset | \hat{a}_i(\tau) \hat{a}_i^\dagger(0) | \emptyset \rangle = e^{\mu\tau} \sum_{\nu} \langle \emptyset | \hat{a}_i(\tau) | \nu \rangle \langle \nu | \hat{a}_i^\dagger(0) | \emptyset \rangle \\ &= e^{\mu\tau} \sum_{\nu} \langle \emptyset | e^{\hat{H}\tau} \hat{a}_i e^{-\hat{H}\tau} | \nu \rangle \langle \nu | \hat{a}_i^\dagger | \emptyset \rangle \\ &= \sum_{\nu} \left| \langle \nu | \hat{a}_i^\dagger | \emptyset \rangle \right|^2 e^{-(E_{\nu}-\mu)\tau} \xrightarrow[\tau \rightarrow \infty]{\mu < E_{GS}} Z_{GS}(i) e^{-(E_{GS}-\mu)\tau} \end{aligned}$$

Example 3: Two-Level-System

- Input: $\epsilon = 0.0$, $\Delta = -0.15$, $\mu = -0.2$
- Analytic: $E_{GS} = -0.15$, $Z_{GS}(1) = Z_{GS}(2) = 0.5$
- Fit: $E_{GS} = -0.1499$, $Z_{GS}(1) = Z_{GS}(2) = 0.5021$



Example 4: Fröhlich polaron

With the knowledge of the previous examples it is only a small step to a DiagMC implementation for the Fröhlich polaron problem:

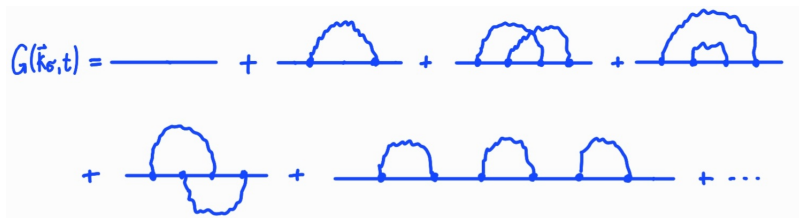
$$\hat{H}_{pol} = \sum_{\mathbf{k}} k^2 \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} + \sum_{\mathbf{q}} \hat{b}_{\mathbf{q}}^\dagger \hat{b}_{\mathbf{q}} + \sum_{\mathbf{k}, \mathbf{q}} i s(\mathbf{q}) \sqrt{\frac{4\pi\alpha}{V}} \frac{1}{q} \hat{a}_{\mathbf{k}+\mathbf{q}}^\dagger \hat{a}_{\mathbf{k}} (\hat{b}_{\mathbf{q}} + \hat{b}_{-\mathbf{q}}^\dagger)$$

$$Q(\{\mathbf{k}, \tau\}) = e^{\mu\tau} G(\mathbf{k}, \tau) = e^{\mu\tau} \left\langle \emptyset \left| \hat{a}_{\mathbf{k}}(\tau) \hat{a}_{\mathbf{k}}^\dagger(0) \right| \emptyset \right\rangle$$

- $\tau \geq 0$
- α is the coupling strength between the electron and phonons (we keep it fixed for each simulation)
- $\mathbf{k}$ is the polaron momentum (we keep it fixed for each simulation)
- μ is an artificial chemical potential (simulation parameter)
- τ_{max} is the max. allowed τ value (simulation parameter)

Example 4: Fröhlich polaron

Diagrammatic series:



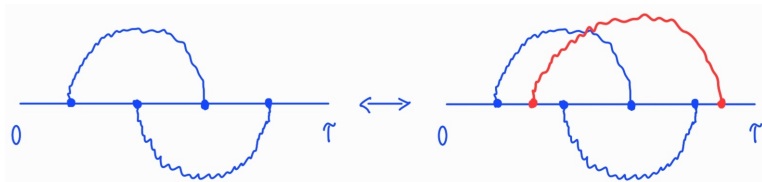
$$e^{\mu\tau} G(\mathbf{k}, \tau) = \sum_{n=0}^{\infty} \sum_{\xi_n} \int_0^{\tau} d\tau_n \cdots \int_0^{\tau_{i+1}} d\tau_i \cdots \int_0^{\tau_2} d\tau_1 \int \cdots \int d\mathbf{q}_1 \dots d\mathbf{q}_{n/2} \\ \times \mathcal{D}_n^{\xi_n}(\mathbf{k}, \tau, \mu; \tau_1, \dots, \tau_n, \mathbf{q}_1, \dots, \mathbf{q}_{n/2})$$

where the diagrams $\mathcal{D}_n^{\xi_n}(\mathbf{k}, \tau, \mu; \tau_1, \dots, \tau_n, \mathbf{q}_1, \dots, \mathbf{q}_{n/2})$ consist of electron propagators $e^{-(k_i^2 - \mu)(\tau_j - \tau_i)}$, phonon propagators $e^{-(\tau_j - \tau_i)}$ squared matrix elements $|V_{\mathbf{q}}|^2$ and a factor of $V/(2\pi)^3$ per wave vector integration.

Example 4: Fröhlich polaron

Updates:

- change- τ update:
 - ▶ same as we had before
- add/remove-vertex updates:



- change-vertex- τ update
- rescale-diagram update
- change-phonon-wave-vector-modulus update
- change-phonon-wave-vector-direction update
- change-topology update

Example 4: Fröhlich polaron

Measurements:

- histogram $H(\mathbf{k}, \tau_i)$ for $G(\mathbf{k}, \tau)$
 - ▶ procedure is exactly the same as before
 - ▶ normalization: use 0th-order diagram
- update statistics
- diagram statistics

Post processing:

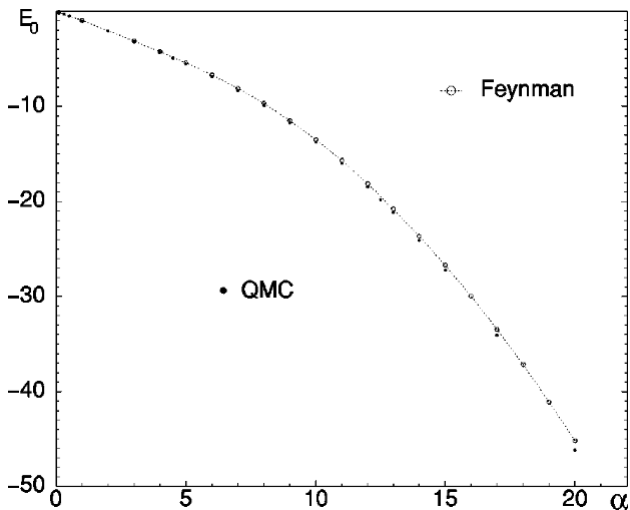
- extract ground state energy $E_{GS}(\mathbf{k})$ and quasiparticle weight $Z_{GS}(\mathbf{k})$ from Green's function:

$$e^{\mu\tau} G(\mathbf{k}, \tau) = e^{\mu\tau} \left\langle \emptyset \left| \hat{a}_{\mathbf{k}}(\tau) \hat{a}_{\mathbf{k}}^\dagger(0) \right| \emptyset \right\rangle \xrightarrow[\tau \rightarrow \infty]{\mu < E_{\nu}} Z_{GS}(\mathbf{k}) e^{-(E_{GS}(\mathbf{k}) - \mu)\tau}$$

Other interesting quantities:

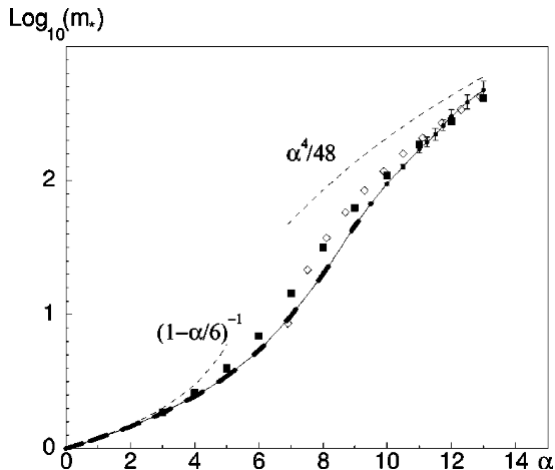
- polaron dispersion, polaron effective mass, structure of phonon cloud, spectral function and optical conductivity for excited spectrum (numerical analytic continuation)

Example 4: Fröhlich polaron



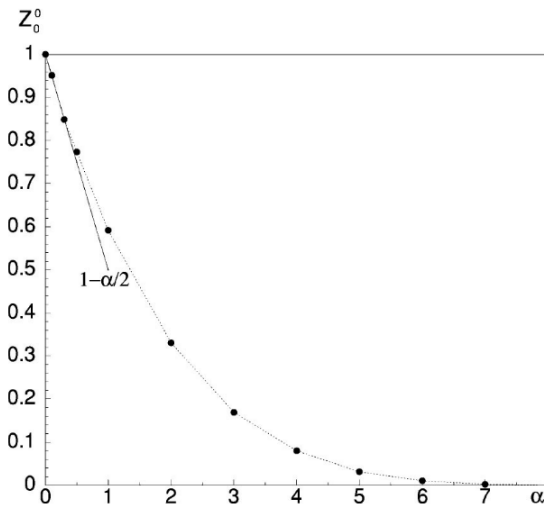
Mishchenko, A. S., et al. "Diagrammatic quantum Monte Carlo study of the Fröhlich polaron." *Physical Review B* 62.10 (2000): 6317.

Example 4: Fröhlich polaron



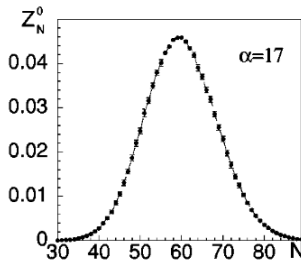
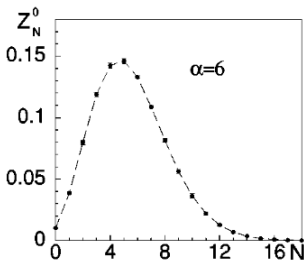
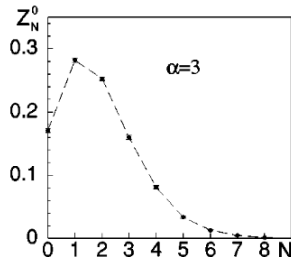
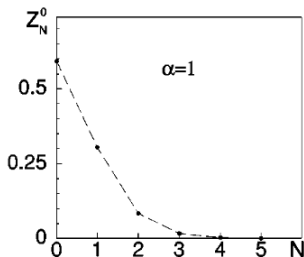
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Example 4: Fröhlich polaron



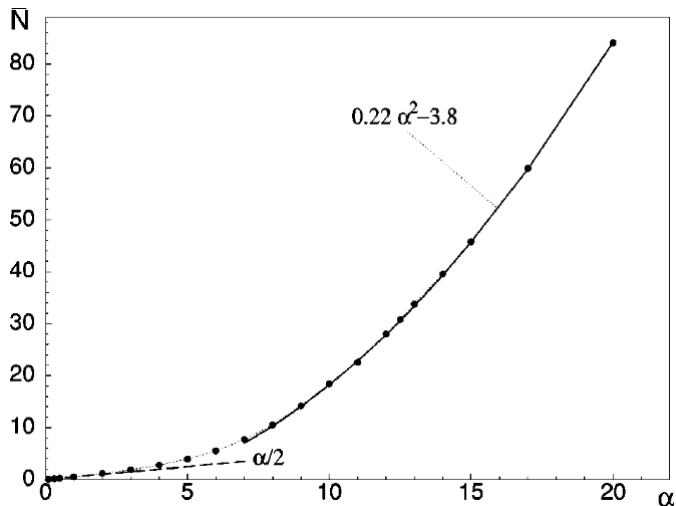
Mishchenko, A. S., et al. "Diagrammatic quantum Monte Carlo study of the Fröhlich polaron." *Physical Review B* 62.10 (2000): 6317.

Example 4: Fröhlich polaron



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Example 4: Fröhlich polaron



Mishchenko, A. S., et al. "Diagrammatic quantum Monte Carlo study of the Fröhlich polaron." *Physical Review B* 62.10 (2000): 6317.

The end